MAT2002 **Name (Print):** _____
Spring 2024
Final Exam
15/05/2024
Time Limit: 120 Minutes

	$f(t) = \mathcal{L}^{-1}[F(s)]$	$F(s) = \mathcal{L}[f(t)]$
1.	1	$\frac{1}{s}, \quad s > 0$
2.	e^{at}	$\frac{1}{s-a}, \quad s > a$
3.	$t^n, \quad n = \text{positive integer}$	$\frac{n!}{s^{n+1}}, \quad s > 0$
4.	$t^p, \quad p > -1$	$\frac{\Gamma(p+1)}{s^{p+1}}, \quad s > 0$
5.	$\sin at$	$\frac{a}{s^2+a^2}, \quad s > 0$
6.	$\cos at$	$\frac{s}{s^2+a^2}, \quad s > 0$
7.	$\sinh at$	$\frac{a}{s^2-a^2}, \quad s > \lvert a \rvert$
8.	$\cosh at$	$\frac{s}{s^2-a^2}, \quad s > \lvert a \rvert$
9.	$e^{at} \sin bt$	$\frac{b}{(s-a)^2+b^2}, \quad s > a$
10.	$e^{at} \cos bt$	$\frac{s-a}{(s-a)^2+b^2}, \quad s > a$
11.	$t^n e^{at}, \quad n = \text{positive integer}$	$\frac{n!}{(s-a)^{n+1}}, \quad s > a$
12.	$u_c(t)$	$\frac{e^{-cs}}{s}, \quad s > 0$
13.	$u_c(t) f(t-c)$	$e^{-cs} F(s)$
14.	$e^{ct} f(t)$	$F(s-c)$
15.	$f(ct)$	$\frac{1}{c} F\left(\frac{s}{c}\right), \quad c > 0$
17.	$\delta(t-c)$	e^{-cs}
18.	$f^{(n)}(t)$	$s^n F(s) - s^{n-1} f(0) - \cdots - f^{(n-1)}(0)$
18a.	$f'(t)$	$sF(s) - f(0)$
18b.	$f''(t)$	$s^2 F(s) - sf(0) - f'(0)$
19.	$(-t)^n f(t)$	$F^{(n)}(s)$

Figure 1: Elementary Laplace transforms.

1. (30') True or false?
   (a) The equation $5y' + 2xy^2 = 1$ is a linear differential equation.
   (b) If $y' = y + 2t$, then $0 = y + 2t$ is an equilibrium solution.
   (c) $f * 1 = 1 * f = f$, where $*$ is the convolution operator.
   (d) For the Dirac delta function $\delta(t)$,
   $$\int_{-\infty}^{\infty} \delta\left(t - \frac{\pi}{3}\right) \sin t \, \mathrm{d}t = \frac{\sqrt{3}}{2}.$$
   (e) Assume that the characteristic roots of a linear homogeneous system are all distinct real numbers. Then the system has a fundamental set of exponential solutions.
   (f) Let $\lambda = a + bi$ be a complex characteristic root of a linear homogeneous system of differential equations and $\mathbf{v} = \mathbf{v}_R + i\mathbf{v}_I$ an eigenvector associated to $\lambda, \mathbf{v}_R, \mathbf{v}_I \in \mathbb{R}^2$. Then $e^{at}\left(\sin(bt)\mathbf{v}_R + \cos(bt)\mathbf{v}_I\right)$ is a solution to the system.
   (g) There exists a matrix A such that $e^A = \begin{pmatrix} 1 & -1 & 2 \\ 1 & 3 & -4 \\ 1 & -1 & 2 \end{pmatrix}$.
   (h) The critical point $(4, 3)$ of the system given below is an asymptotically stable spiral point
   $$\begin{cases} x' = 33 - 10x - 3y + x^2 \\ y' = -18 + 6x + 2y - xy. \end{cases}$$
   (i) The system of equations $x' = 2x - 3x^2 - 2xy;\ y' = y - 0.5y^2 - 3xy$ describes a competitive species model. This system has four critical points, including two nodal points and two saddle points.
   (j) Let $\mathbf{x}' = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mathbf{x}$ with real constants $a,\ b,\ c,\ d$. If $b > 0$ and $c > 0$, then the critical point is not a spiral.

Solution: (a) False (b) False (c) False (d) True (e) True (f) True (g) False (h) True (i) True (j) True

2. (8') Solve the differential equation
   $$(2xy + y^3)dx + (x^2 + 3xy^2 - 2y)dy = 0.$$

Solution: Exact equation.
$$x^2y + xy^3 - y^2 = C.$$

3. (12') Solve the linear system:
   $$\mathbf{y}' = \mathbf{A}\mathbf{y}, \quad \mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{1}$$

Solution: SN decomposition method or use generalized eigenvector $\mathbf{r}_0^{(i)}$.

$$\mathbf{y}(t) = c_1\mathbf{y}_1 + c_2\mathbf{y}_2 + c_3\mathbf{y}_3 = c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^t + c_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} e^{2t} + c_3 \begin{pmatrix} 0 \\ t \\ 1 \end{pmatrix} e^{2t}. \tag{2}$$

Note that the solution's form might be slightly different.

4. (15') Find the general real-valued solution of the 2×2 system

$$\mathbf{x}' = \begin{pmatrix} 1 & 2 \\ -5 & -1 \end{pmatrix} \mathbf{x}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}. \tag{3}$$

Determine the type and stability of the critical point $(0, 0)$, and draw the phase portrait for this linear system.

Solution: The eigenvalues of $\begin{pmatrix} 1 & 2 \\ -5 & -1 \end{pmatrix}$ are

$$r_{1,2} = \pm 3i.$$

The corresponding eigenvectors are

$$\xi^{(1)} = \begin{pmatrix} -2 \\ 1 - 3i \end{pmatrix}, \quad \xi^{(2)} = \begin{pmatrix} -2 \\ 1 + 3i \end{pmatrix}.$$

The general complex-valued solution is

$$\mathbf{x}(t) = d_1 \begin{pmatrix} -2 \\ 1 - 3i \end{pmatrix} e^{3i} + d_2 \begin{pmatrix} -2 \\ 1 + 3i \end{pmatrix} e^{-3i}.$$

The general real-valued solution is

$$\mathbf{x}(t) = c_1 \begin{pmatrix} -2\cos(3t) \\ \cos(3t) + 3\sin(3t) \end{pmatrix} + c_2 \begin{pmatrix} -2\sin(3t) \\ \sin(3t) - 3\cos(3t) \end{pmatrix}.$$

Center, stable critical point.

5. (20') Consider the function

$$f(t) = \begin{cases} 0, & 0 \le t < \dfrac{\pi}{2}, \\ \sin t, & \dfrac{\pi}{2} \le t < \pi, \\ 0, & t \ge \pi. \end{cases}$$

(a) Rewrite $f(t)$ in terms of unit step functions.

(b) Prove the following Laplace transform:

$$\mathcal{L}\{u_c(t)f(t-c)\} = e^{-cs}F(s), \tag{4}$$

where u_c is the unit step function with $c > 0$ and $F(s)$ is the Laplace transform of $f(t)$.

(c) Compute $\mathcal{L}\{f(t)\}$.

(d) Use the results in (c) to solve the initial value problem

$$y'' + 4y = f(t), \quad y(0) = 0,\ y'(0) = 0.$$

Solution:

(a)

$$f(t) = u_{\frac{\pi}{2}}(t)\sin t - u_\pi(t)\sin t.$$

(b) Skipped.

(c)

$$\begin{aligned} F(s) = \mathcal{L}\{f(t)\} &= \mathcal{L}\left\{u_{\frac{\pi}{2}}(t)\sin\left(t+\frac{\pi}{2}-\frac{\pi}{2}\right) - u_\pi(t)\sin\left(t+\pi-\pi\right)\right\} \\ &= e^{-\frac{\pi}{2}s}\frac{s}{s^2+1} - e^{-\pi s}\frac{-1}{s^2+1} \\ &= \frac{e^{-\frac{\pi}{2}s}s + e^{-\pi s}}{s^2+1}. \end{aligned}$$

(d) Taking the Laplace transform, we have

$$(s^2+4)Y(s) = F(s).$$

Hence, we have

$$\begin{aligned} y(t) = \mathcal{L}^{-1}\{Y(s)\} &= \mathcal{L}^{-1}\left\{\frac{e^{-\frac{\pi}{2}s}s + e^{-\pi s}}{3}\left(\frac{1}{s^2+1} - \frac{1}{s^2+4}\right)\right\} \\ &= \frac{1}{3}\left(u_{\frac{\pi}{2}}(t)\sin t - u_\pi(t)\sin t\right) - \frac{1}{3}\left(-u_{\frac{\pi}{2}}(t)\cos(2t) + \frac{1}{2}u_\pi(t)\sin(2t)\right) \\ &= \frac{1}{3}\left(u_{\frac{\pi}{2}}(t) - u_\pi(t)\right)\sin t + \frac{1}{3}u_{\frac{\pi}{2}}(t)\cos(2t) - \frac{1}{6}u_\pi(t)\sin(2t). \end{aligned}$$

Please confirm the above answers are correct.

6. (15') Consider the system of differential equations

$$\begin{cases} x' = x - t^2 + 2t \\ y' = 5x + 2y - 5t^2 - 2t + 1 \end{cases}$$

(a) Determine whether the equilibrium point of the homogeneous system is unstable, stable or asymptotically stable.

(b) Find the general solution to the homogeneous system.

(c) Find the general solution to the nonhomogeneous system.

Solution:

(a) Homogeneous system:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 5 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Eigenvalues $r_1,\ r_2 > 0$. Unstable.

(b)

$$\begin{pmatrix} x \\ y \end{pmatrix} = C_1 e^{2t} \begin{pmatrix} 0 \\ 1 \end{pmatrix} + C_2 e^{t} \begin{pmatrix} 1 \\ -5 \end{pmatrix}.$$

(c)

$$\begin{pmatrix} x \\ y \end{pmatrix} = C_1 e^{2t} \begin{pmatrix} 0 \\ 1 \end{pmatrix} + C_2 e^{t} \begin{pmatrix} 1 \\ -5 \end{pmatrix} + \begin{pmatrix} t^2 \\ t \end{pmatrix}.$$